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ABSTRACT

The Port Stanley till, located along the north shore of Lake Erie, was sampled in detail at vertical intervals of 25 cm at one site near Wheatley, Ontario (19 blocks, 90 specimens), and at six other sites (39 blocks, 235 specimens) along 100 km of the shore of Lake Erie east of Wheatley to the Port Stanley area. The detrital remanent magnetization (DRM) is moderately intense (mean $J_{NRM} \approx 4.1 \times 10^{-5}$ emu per cm^3) and stable (mean coercivity ≈300 Oe). The DRM mean directions after alternating-field (AF) demagnetization for each site are well grouped (mean α_{95} = 8.6°), indicating a mean direction by sites of 7.4°, 47.2° (R = 6.9830, α_{95} = 3.2°) and a mean virtual geomagnetic pole (VGP) position of lat 74.7° N., long 78.2° E. (δ_m = 4.2°, δ_p = 2.7°). The paleomagnetic data suggest a rapid deposition rate for the Port Stanley till of 100 to 400 yr, in reasonable agreement with the minimum geologic estimate of about 500 yr. Anisotropy of magnetic susceptibility measurements made at each site provides a reliable estimate of the fabric in the till.

INTRODUCTION

A magnetic study of the Port Stanley till near Port Alma, Ontario (Fig. 1, site 2), on the north shore of Lake Erie revealed that the till has a strong characteristic magnetism (RM_c) (Gravenor and others, 1973). The RM_c was probably developed when the magnetic particles were released from the base of the ice through a water or slurry layer, thus forming a deposit having detrital remanent magnetization (DRM). A detailed study of the anisotropy of magnetic susceptibility (Gravenor and others, 1973) ruled out the possibility that the alignment of the magnetic particles to form the RM_c was caused by glacier movement. The purpose of the present study is to examine the Port Stanley till at additional sites along the north shore of Lake Erie to determine if the till sheet exhibits a uniform remanent magnetization direction in lateral extent and thereby determine if a till sheet can be characterized by its virtual geomagnetic pole (VGP) position.

Seven sites were investigated on the north shore of Lake Erie between Wheatley and J. E. Pearce Park near Port Stanley, some 100 km to the east (Fig. 1). Examination of the till at 15 intermediate locations along the 100 km of shoreline indicated that a single till sheet was sampled. A granulometric analysis of 2,000 g of the Port Stanley till taken from the Port Alma area showed that it is composed of about 44 percent clay, 28 percent silt, 20 percent sand, and the remaining 8 percent coarse sand (>1 mm) and gravel. Where sampled, the till is massive in character, gray in color, and lacks evidence of stratification or interlayered stratified sediments. The Port Stanley till is of late Wisconsin age (de Vries and Dreimanis, 1960).

SAMPLING AND MEASUREMENT PROCEDURE

At each of the seven sites, from 4 to 19 oriented cubic blocks of till measuring about 15 cm on each side were cut from the cliffs in a vertical sequence at about 25 cm intervals. The thickness of the sampled sections ranged from 10 to 20 m. The blocks were oriented with a Brunton compass and level. An average of 5 cores were drilled from each block using a diamond bit in a drill press. The cores were then trimmed to specimen size (2.5 cm in diameter and 2.5 cm in length), labelled, and coated with polyurethane paint. Initially, the natural remanent magnetization (NRM) of one specimen from each site was measured on a Schonstedt SSM–A spinner magnetometer. The remanence of each specimen was remeasured after alternating-field (AF) demagnetization using a Schonstedt GSD–1 demagnetizer set at step increments ranging from 50 to 200 Oe and up to a maximum intensity of between 500 and 800 Oe (Fig. 2). All specimens reacted similarly to demagnetization, confirming the presence in the till at all sites of about 10^5 magnetite particles of <37μ size carrying the DRM as found in site 2 (Gravenor and others, 1973). Their remanence intensity (Fig. 2) decreased nearly linearly with increasing AF intensity, giving a mean coercivity of 306 Oe (in other words, AF intensity required to reduce the NRM intensity by one half) with a standard deviation of ±23 Oe. Their remanence directions typically show (1) a few degrees (≈4°) of movement away from the Earth's present magnetic field direction at 50 Oe, which presumably reflects the removal of isothermal remanent magnetization (IRM)

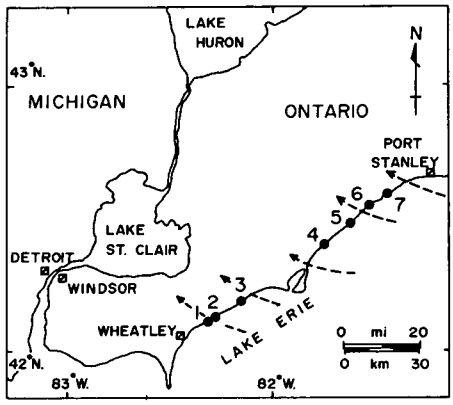


Figure 1. Sampling sites (as solid circles) for the Port Stanley till. The dashed lines indicate the last direction of ice flow (Chapman and Putnam, 1951; Dreimanis, 1961). The following is the location (by latitude and longitude) of the sampled sites: (1) 42° 09′ N., 82° 20′ W.; (2) 42° 10′ N., 82° 19′ W.; (3) 42° 13′ N., 82° 10′ W.; (4) 42° 26′ N., 81° 43′ W.; (5) 42° 29′ N., 81° 38′ W.; (6) 42° 33′ N., 81° 33′ W.; and (7) 42° 36′ N., 81° 27′ W.

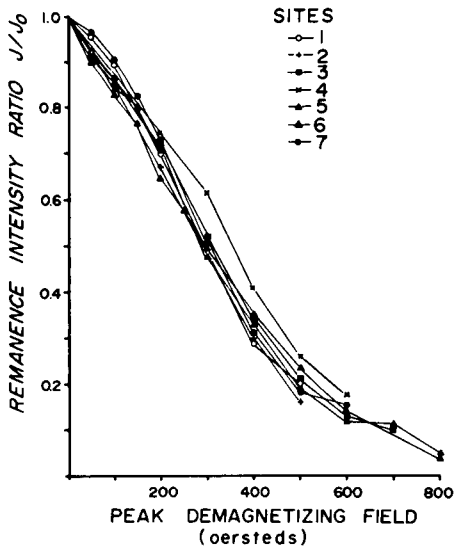


Figure 2. AF demagnetization curve for one specimen from each site; the remanence intensity ratio J/J_0 is plotted against the demagnetizing field. The J_0 values for the seven specimens are 2.33, 2.21, 3.95, 1.82, 4.67, 4.96, and 3.95 × 10^{-5} emu per cm^3, respectively.

components; (2) a clustering between 50 and 400 Oe, which is thought to be the characteristic remanent magnetization (RM_c); and (3) an increasing scatter about the cluster above 400 Oe as anhysteretic remanence components (ARM) are induced in the specimens by the demagnetizing

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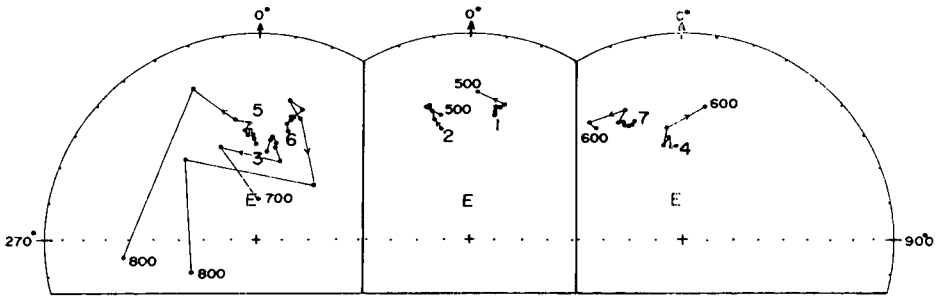


Figure 3. AF demagnetization paths for one specimen from each site plotted on a Schmidt stereonet. The numbers 1 to 7 indicate the site and are beside its NRM direction. The maximum AF intensity employed (500 to 800 Oe) is beside the final direction. The step intervals are given in Figure 2. E is centered on the Earth's present magnetic field direction in the sampling area.

TABLE 1. SUMMARY OF REMANENT MAGNETIZATION DATA

Site	Number of Blocks	Length of Vector Resultant R	Mean Remanence Direction: Declination (degrees)	Inclination + down (degrees)	Precision Parameter K	α_{95} (degrees)	Virtual North Pole Position: Longitude East (degrees)	Latitude North (degrees)	δ_p (degrees)	δ_m (degrees)
1	14	13.8597	2.9	48.0	93	4.2	86.5	76.7	3.5	5.4
2	19	18.6933	9.6	42.6	59	4.4	71.0	70.9	3.4	5.5
3	5	4.9527	3.7	46.7	85	8.4	84.3	75.5	6.9	10.8
4	5	4.9777	6.4	44.1	179	5.7	77.8	72.8	4.5	7.2
5	4	3.9709	7.1	48.0	103	9.1	72.8	75.5	7.8	11.9
6	6	5.8815	15.7	47.5	42	10.4	51.0	71.2	3.8	13.6
7	5	4.7941	6.2	52.6	19	17.8	69.2	79.4	16.9	24.5
Mean: By Site	7	6.9830	7.4	47.2	353	3.2	72.8	74.7	2.7	4.2

Notes: α_{95} radius of cone of 95% confidence
δ_p, δ_m semi-axes of the oval of 95% confidence along and perpendicular to the site-pole great circle.

process (Fig. 3). Thus, on the basis of these cleaning tests, the remaining specimens were AF demagnetized at 200 Oe and their RM_c measured. Assigning the RM_c direction of each specimen unit weight, the mean direction (RM_b) was calculated for each block. The RM_b directions were averaged to calculate the mean remanence direction for each site (RM_s) along with its precision parameter (K) and radius of the cone of 95 per cent confidence (α_{95}; see Fisher, 1953; Table 1). Similarly, the mean direction for the till sheet (RM_m) was calculated by averaging the RM_s directions for the seven sites. The RM_m direction was used to compute the VGP position and semiaxes (δ_p, δ_m) of the oval of 95 percent confidence about the pole.

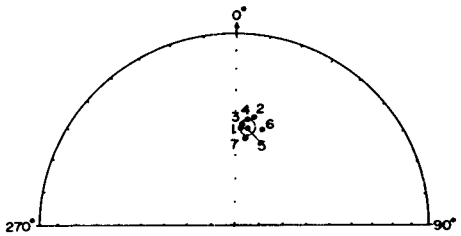


Figure 4. Site mean remanence directions (solid circles) for the Port Stanley till. The mean position of the site means is directly under site 5 and is surrounded by its circle of 95 percent confidence.

DISCUSSION OF THE REMANENCE RESULTS

The within-site variations in the RM_b directions are small for six of the seven sites; therefore, most of the RM_s directions have high K and small α_{95} values (Table 1). The between-site variations in the RM_s directions are also small (Fig. 4), resulting in a direction RM_m for the Port Stanley till sheet having a high K and low δ_{95} value. Table 2 gives the results of a test of the angular variance ratio (Larochelle, 1968) and shows a δ_b^2/δ_w^2 ratio of 8.66, which is greater than the statistic $F_{2(B-1),\ 2(N-B),\ 0.05}$. Thus, the results of this test indicate that there are good reasons to assume that the site mean directions are significantly distinct.

It has been suggested (Gravenor and others, 1973) that the within-site variations in the RM_b directions primarily reflect the variation in the direction of the Earth's magnetic field at each site over a few decades during deposition of the till. Similarly, it is reasonable to assume that the between-site variation in the RM_s direction results primarily from the variation in the time of deposition of the till at each site. In short, both the within- and between-site variations probably arise from the time-transgressive nature of the process of till deposition in the vertical and lateral directions.

The VGP, calculated from the RM_m direction using the dipole hypothesis, is just off the northern coast of central Russia (Fig. 5). The clustering of the virtual geomagnetic poles about the mean VGP for the till sheet is good and indicates that the sampled part of the till sheet, extending laterally some 100 km and vertically 10 m, was deposited in the relatively short time interval of several hundred years. This is consistent with the available radiometric dates and other geologic evidence for the short duration (400 to 500 yr) of the deposition of the Port Stanley till (Goldthwait and others, 1965, p. 94; Gravenor and others, 1973). The within- and between-site variations in the intensity of the RM of the till after demagnetization at a peak AF intensity of 200 Oe are small, resulting in a mean intensity for the till sheet of 2.91×10^{-5} emu per cm^3 with a relatively small standard deviation ($\pm .44 \times 10^{-5}$ emu per cm^3). This confirms the constancy of the assemblage of magnetic minerals within and between sites in the Port Stanley till.

MAGNETIC SUSCEPTIBILITY ANISOTROPY

Fuller (1964) showed that measuring the magnetic susceptibility anisotropy provided a rapid method for till-fabric determination. He found that the maximum susceptibility direction in the till was similar to the orientation of the long axes of contained clasts. The magnetic susceptibility anisotropy was measured for 10 to 24 of the

Table 2: Angular Variance Analysis

B	N	δ_w^2, radians^2	δ_b^2, radians^2	δ_b^2/δ_w^2	$F_{2(B-1),\ 2(N-B),\ 0.05}$
7	58	.005544	.04802	8.6616	2.36

Notes: B number of sites
N number of blocks
δ_w^2 and δ_b^2 within-site and between-site angular variances

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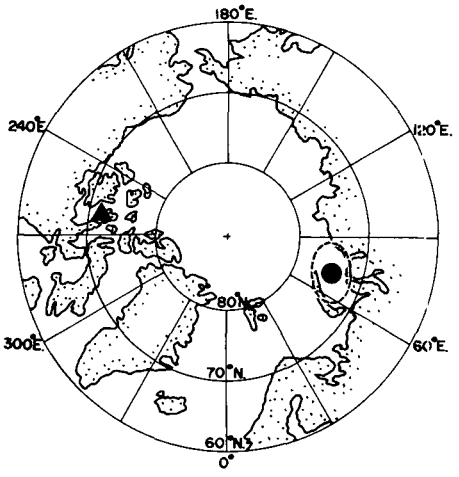


Figure 5. Polar projection of the northern hemisphere above lat 60° N. showing the VGP position (solid circle) for the Port Stanley till with its oval of 95 percent confidence (dashed line), and the Earth's present geomagnetic pole (solid triangle).

specimens from each site on a Schonstedt SSM–1A spinner magnetometer utilizing a 6-spin method. The bulk susceptibility for the same specimens was measured on a transformer bridge (Christie and Symons, 1969). For each specimen, the magnitude and direction of the minimum, intermediate, and maximum susceptibility were calculated by the method of successive approximations (Nye, 1960). The individual values were then averaged for each site, and the results are illustrated in Figure 6 and summarized in Table 3. The directions of the minimum susceptibility are tightly clustered close to the vertical direction. The plane containing the maximum and intermediate susceptibility is nearly horizontal, and the clustering of the maximum and intermediate susceptibility directions is not as tight as for the minimum susceptibility directions. This is typical of fabrics in which the magnitude of the maximum and intermediate susceptibilities differ by about ten percent, however (Stone, 1963).

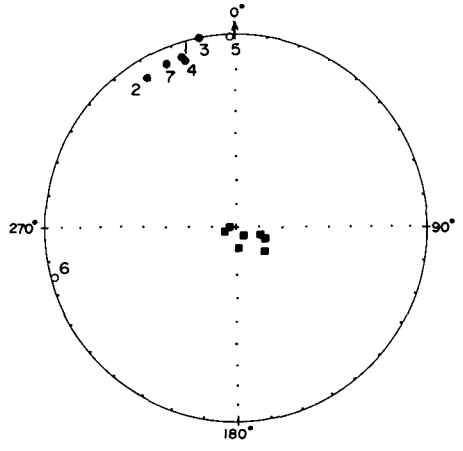


Figure 6. Anisotropy of magnetic susceptibility data showing the mean directions of maximum susceptibility as circles with their site number and of minimum susceptibility as squares. Solid (open) symbols indicate +ve (−ve) or downward (upward) inclinations on the stereonet.

TABLE 3. SUMMARY OF ANISOTROPY OF MAGNETIC SUSCEPTIBILITY DATA

Site	Number of specimens	Maximum susceptibility direction			Minimum susceptibility direction			Mean magnitude ratios	
		Declination (degrees)	Inclination (degrees)	α_{95} (degrees)	Declination (degrees)	Inclination (degrees)	α_{95} (degrees)	Max/Int	Min/Int
1	18	342.1	8.9	19.7	118.9	76.7	7.5	1.024	0.922
2	24	329.9	12.4	18.0	138.1	85.2	3.0	1.021	0.908
3	10	349.3	1.3	6.6	275.5	87.5	3.9	1.045	0.843
4	16	342.9	11.4	12.2	105.1	78.1	3.6	1.043	0.919
5	14	358.6	-1.1	31.0	242.3	85.5	5.1	1.016	0.921
6	14	255.4	-3.0	29.5	172.7	82.3	4.5	1.025	0.830
7	19	337.0	9.3	22.0	128.5	73.4	4.6	1.020	0.883

The direction of movement of the Lake Erie lobe during deposition of the Port Stanley till is known from the long axes of end moraines found north of the Lake Erie basin in southwestern Ontario and from pebble-fabric studies (Chapman and Putnam, 1951; Dreimanis and Reavely, 1953; Dreimanis, 1961). Dreimanis and Reavely (1953, p. 244) showed that, during the early stages of deposition of the Port Stanley till, the glacier flowed northwest. Throughout most of the middle of the deposition period, the ice flowed more or less westerly or parallel to Lake Erie. During the latter stages, the ice again flowed north-northwest as shown in Figure 1. Except for site 6, the site mean maximum susceptibility directions are north-northwest and appear to reflect the local ice-flow direction during the latter stages of deposition of the till. These local site directions reflect the general regional trend of ice flow, suggesting that localized deviations of ice flow at individual sites in this till are minimal. The westerly direction of the maximum susceptibility obtained at site 6 may represent a phase of the till deposited when the till was flowing parallel to the Lake Erie basin and is representative of the middle deposition period given by Dreimanis and Reavely (1953, p. 249). Thus, in general, the maximum magnetic susceptibility direction for the Port Stanley till apparently provides a record of the ice-flow direction. The question that still remains is whether the magnetic particles giving rise to the magnetic susceptibility anisotropy are preferentially oriented while in the flowing ice and retain the orientation during deposition from stagnant ice as suggested by Harrison (1957) or whether the orientation takes place at the ice-till interface during deposition from actively flowing ice. The change in the ice-flow direction during the deposition of the Port Stanley till reported by Dreimanis and Reavely (1953) would suggest that this till was deposited from actively flowing ice.

Our recent unpublished laboratory studies on artificial till slurries show that the degree of alignment of magnetite particles in the Earth's field is strongly dependent on the size of the magnetite. The results show that the fine magnetite particles ($<37\mu$) in clayey till are free to align themselves in the Earth's field whereas the coarse material ($>37\mu$) is not significantly aligned. Translating these results to the behavior of magnetite in natural till, it is probable that the DRM is caused by alignment of the fine magnetite particles in the Earth's field at the time of deposition, and the coarse magnetite retains the original alignment created by the ice flow and hence gives rise to the magnetic susceptibility anisotropy.

CONCLUSIONS

There are four substantive conclusions to be drawn from this study:

1. A part of the Port Stanley till sheet, extending some 100 km laterally and 10 m vertically, retains a stable remanent magnetization of DRM origin with consistent magnetic intensity, direction, and coercivity spectrum throughout.

2. The till sheet retains a uniform $\mathbf{RM}_c$ direction after AF demagnetization; therefore, it can be characterized by its mean VGP position of lat 74.7° N., long 72.8° E. (δ_m = 4.2°, δ_p = 2.7°) for purposes of dating and correlation.

3. From tests of the angular variance ratio, it is clear that each site is a statistically independent observation which reflects the time-transgressive nature of the till deposition. It is apparent from the available radiocarbon dates and the vertical section sampled at site 2 (Gravenor and others, 1973) that deposition must have occurred rapidly during part of a secular-variation cycle or during a time interval of 300 ± 200 yr.

4. The anisotropy of magnetic susceptibility directions apparently reflects the ice-flow direction during deposition of the till and is unrelated to the DRM.

Finally, there are two aspects of this study that clearly require further work. First, the problem of "inclination error" in the $\mathbf{RM}_c$ direction in tills is of critical importance. Our preliminary laboratory studies indicate that this may not be a serious source of error. Second, the necessity for a reliable secular-variation pole path for the Pleistocene Epoch is now evident if a reliable paleomagnetic geochronology is to be established.

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